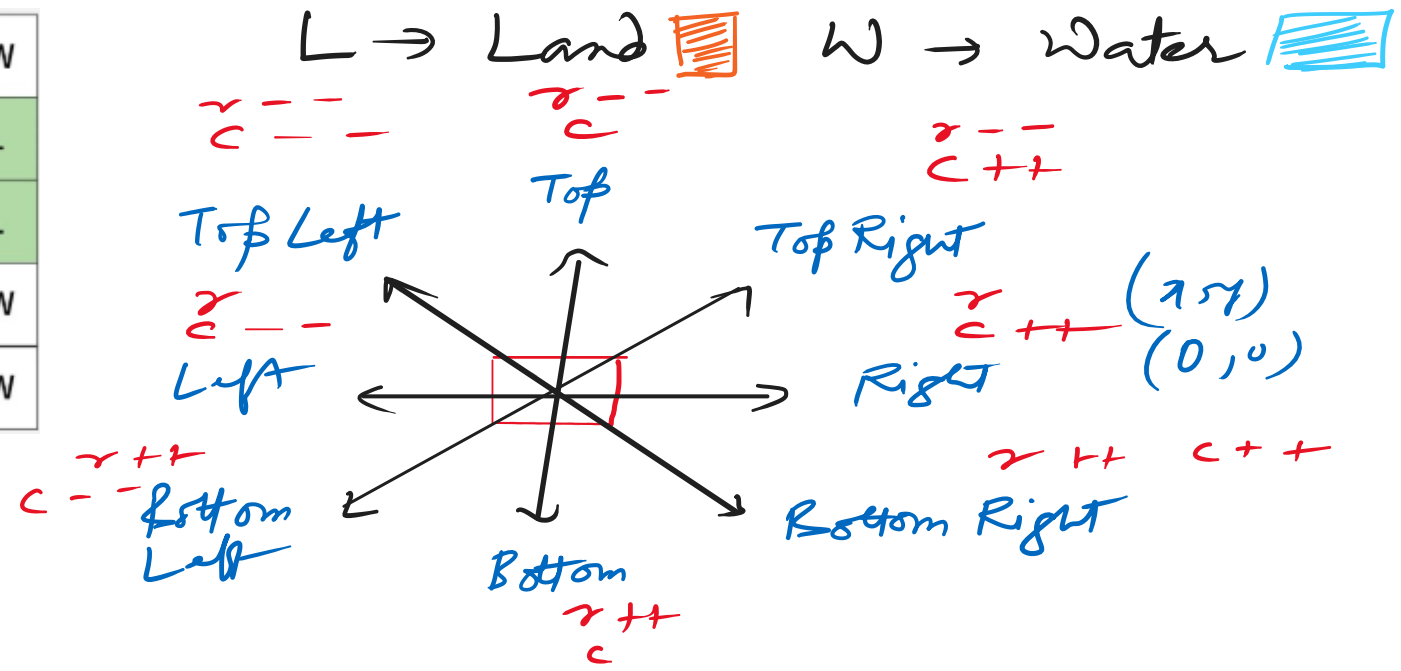


Islands For Islands $\Rightarrow$ Number of Islands

L	L	W	W	W
W	L	W	W	L
L	W	W	L	L
W	W	W	W	W
L	W	L	L	W



Direction	Change in Coordinates	dx	dy
Top Left	$(x-1, y-1)$	-1	-1
Top	$(x-1, y)$	-1	0
Top Right	$(x-1, y+1)$	-1	+1
Left	$(x, y-1)$	0	-1
Right	$(x, y+1)$	0	+1
Bottom Left	$(x+1, y-1)$	+1	-1
Bottom	$(x+1, y)$	+1	0
Bottom Right	$(x+1, y+1)$	+1	+1

The idea is to count the no. of dfs or bfs calls. No. of dfs calls = No. of islands

Disjoint Set Data Structure $\Rightarrow$

Let's consider a graph with 2 components $\Rightarrow$

Component 1 $\Rightarrow$ ① — ② — ③ — ④

Component 2 $\Rightarrow$ ⑤ — ⑥ — ⑦

Time more $\rightarrow O(N+E)$
BFS $\rightarrow$ Same Time (False/No)

Either DFS or BFS is very slow, so we use the Disjoint Set Logic.

Disjoint Set Use Case? $\Rightarrow$ find Parent, union

Check whether two nodes lie in the same component in constant time specifically for dynamically changing graphs.

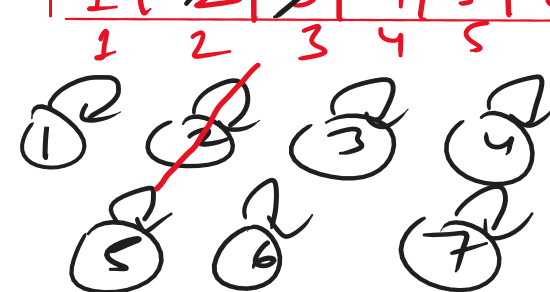
Disjoint Set : Union By Rank

- (u, v)
- $(1, 2)$
- $(2, 3)$
- $(4, 5)$
- $(6, 7)$
- $(5, 6)$
- $(3, 7)$

Initially all nodes are connected to itself as parent

rank $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
1 2 3 4 5 6 7

parent $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
1 2 3 4 5 6 7



$(1, 2)$ $p_u \rightarrow p_1 = 1$
rank $p_u = 0$

Union(u, v) Pseudo Code

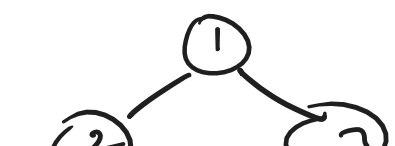
Steps: $\rightarrow$

① Find the ultimate parents of u & v

② Find the rank of p_u, p_v

③ Connect the smaller rank node to the larger rank

rank $p_v = 0$



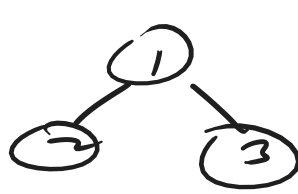
$(2, 3)$ $p_u \rightarrow 2 = 1$
rank = 0

$p_v \rightarrow 3 = 3$
rank = 0

$(4, 5)$ $p_u \rightarrow p_4 = 4$
rank $u = 0$

$p_v \rightarrow p_5 = 5$
rank $5 = 0$

rank $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
1 2 3 4 5 6 7



parent $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
1 2 3 4 5 6 7

$(6, 7)$ $p_u \rightarrow p_6 = 6$

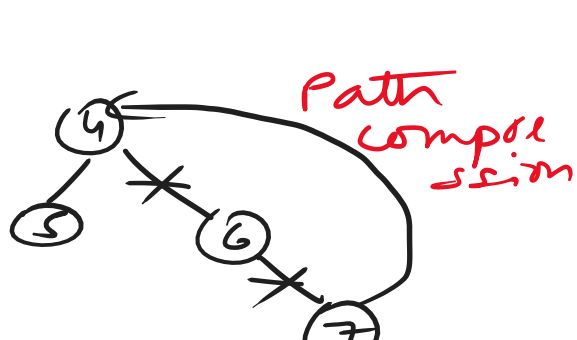
$p_v \rightarrow p_7 = 7$



rank $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
1 2 3 4 5 6 7



parent $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
1 2 3 4 5 6 7



$(5, 6)$ $p_u \rightarrow p_5 = 4$
rank = 1

$p_v = p_6 = 6$
rank = 1

Path Compression $\Rightarrow$

$O(4 \times)$ Constant

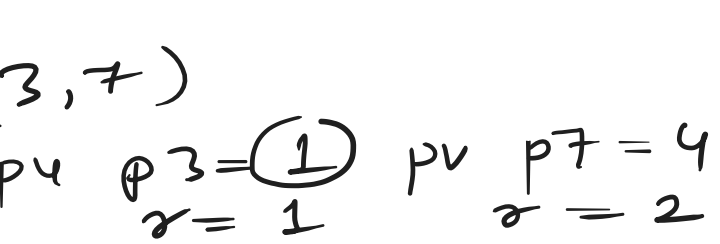
rank $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
1 2 3 4 5 6 7

parent $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
1 2 3 4 5 6 7

Inverse Ackermann $4 \times$

Path Compression

$(3, 7)$ $p_u \rightarrow p_3 = 1$ $p_v \rightarrow p_7 = 4$
rank = 1 rank = 2



Q. Why do we connect a smaller rank node to a larger rank node?

Suppose:

